

E0 232 PROBABILITY AND STATISTICS

MIDTERM 1

Date: 19th September, 2025

Duration: 3 hours

Semester: Hemanth/Fall 2025

Maximum Points: 20

Instructions:

- (1) Please neatly write down the answers on blank sheets of paper. **It is recommended that you write the answer to each question on a new page.**
- (2) Once you are ready to submit, you will need to scan the answer sheets in to a pdf. This pdf will then need to be uploaded to **gradescope**.
- (3) There are **04 problems** in this question paper that you need to answer all of them. Each question is worth **05** points.
- (4) Simply writing down the final answer will not get you the full points. Instead, you must give explicit reasons for each statement that you make in your submission. Your explanation should be good enough to make the TA understand how you arrived at your answer. There are marks for partial solutions.
- (5) Mention all the assumptions you are making. Existing result from any literature can be used directly in your proof; however, please explicitly cite the same.
- (6) Use a calculator when necessary.
- (7) Do not cheat. Any academic misconduct will be dealt with strictly using all permissible laws of the institute.

(1) Let $h : \Omega \rightarrow S$ be a mapping between two sets.

(a) **TRUE or FALSE (justify):** For a σ -field $\mathcal{F}$ on S , the inverse image

$$h^{-1}(\mathcal{F}) := \{ h^{-1}(F) : F \in \mathcal{F} \}$$

is a σ -field on Ω .

(b) **TRUE or FALSE (justify):** For a σ -field $\mathcal{A}$ on Ω , the direct image family

$$h(\mathcal{A}) := \{ h(A) : A \in \mathcal{A} \}$$

is a σ -field on S .

(2) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let X be a general random variable. Suppose $A \in \mathcal{F}$ with $\mathbb{P}(A) = 0$. Is $\mathbb{E}[X \mathbf{1}_A]$ defined? If so, what is its value?

(3) Let $(X_n)_{n \geq 1}$ be random variables on a common probability space defined by

$$X_n = \begin{cases} \frac{1}{n}, & \text{with probability } \frac{1}{n^2}, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Compute $\mathbb{P}(X_n \neq 0)$.

(b) Show that $\sum_{n=1}^{\infty} \mathbb{P}(X_n \neq 0) < \infty$.

(c) Show that the random series $\sum_{n=1}^{\infty} X_n$ converges almost surely.

(4) Let $(X_n)_{n \geq 1}$ be a sequence of independent random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, and consider

$$A := \left\{ \limsup_{n \rightarrow \infty} \frac{X_n}{n} = 0 \right\}.$$

Show that $\mathbb{P}(A) \in \{0, 1\}$.